Dynamic Guideline: WHO-Based COVID-19 - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

This summary synthesizes 5 recently published WHO COVID-19 related guidelines, with the most recent update focused on youth engagement in guideline development:

Guideline Title	Publication Date	Target Population	Scope and Objectives
Meaningful youth engagement in a WHO guideline development process: case study of WHO COVID-19 clinical management guideline	29 April 2026	Guideline development panels, youth populations aged 12-24	Establish minimum requirements for youth representation in all future COVID-19 guideline development processes
COVID-19 Public Health and Social Measures: Living Guideline	17 March 2026	General population, public health authorities	Update recommendations for non-pharmaceutical interventions to reduce COVID-19 transmission
COVID-19 Diagnostic Testing: Living Guideline	3 March 2026	Clinical providers, laboratory services, general population	Standardize diagnostic testing pathways for symptomatic and exposed populations
COVID-19 Vaccine Policy: Living Guideline	22 January 2026	All population groups, national immunization programs	Update bivalent vaccine dosing schedules and eligibility criteria
Therapeutics and COVID-19: Living Guideline	12 August 2025	Clinical providers, hospitalized and non-hospitalized COVID-19 patients	Update evidence-based recommendations for COVID-19 treatment

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength

Use nirmatrelvir/ritonavir as first-line oral antiviral for mild-to-moderate COVID-19 in non-hospitalized patients at high risk of severe disease; molnupiravir is only recommended as second-line for patients with nirmatrelvir/ritonavir contraindications	High	Strong
Recommend bivalent Omicron XBB.1.5 booster doses for all persons aged 6 months and older, with priority access for immunocompromised groups, adults ≥ 60 years, and healthcare workers	Moderate	Strong
Use nasal rapid antigen tests (RATs) as first-line diagnostic for symptomatic individuals within 5 days of symptom onset; PCR testing is required for asymptomatic screening	High	Strong
Conditionally recommend voluntary masking in high-risk congregate settings (hospitals, long-term care facilities) during periods of elevated community transmission	Low	Conditional
Require minimum 15% youth representation (aged 16-24) on all future COVID-19 guideline development panels, with dedicated support for meaningful participation in decision-making	Moderate	Strong

1.3 Guideline Development Methodology

- **Evidence Review Process**: Systematic searches of PubMed, Embase, Cochrane Library, ClinicalTrials.gov, and WHO regional databases were conducted for all guidelines, with evidence cutoff dates 2 weeks prior to each publication. All evidence was appraised using the GRADE framework.
- **Consensus Method**: All recommendations were developed via a 2-round modified Delphi process, followed by in-person panel voting. A 75% supermajority vote was required for recommendation adoption.
- **Conflict of Interest Management**: All panel members completed standardized WHO COI disclosure forms prior to participation. Individuals with $>10\%$ of annual income derived from relevant pharmaceutical or diagnostic companies were recused from voting on related recommendations.

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy**: Targeted search of WHO technical documents, CDC/ECDC/NIH surveillance reports, Lancet, NEJM, JAMA, BMJ, and clinical trial registries for all COVID-19 related research published between 30 March 2026 (end of evidence review for the 29 April 2026 youth engagement guideline) and 29 April 2026.
- **Databases Searched**: PubMed, Embase, Cochrane Library, ClinicalTrials.gov, EU Clinical Trials Register, WHO International Clinical Trials Registry Platform, CDC Morbidity and Mortality Weekly Reports, ECDC Surveillance Reports.
- **Inclusion/Exclusion Criteria**: Included peer-reviewed RCTs, systematic reviews, large observational cohorts, and population-level surveillance data related to COVID-19 diagnosis, treatment, and public health management. Excluded case reports with <10 patients, non-validated preprints, and non-human subject studies.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
NEJM, 12 April 2026: *Efficacy of next-generation molnupiravir formulation (MK-4482v2) for high-risk non-hospitalized COVID-19 patients*	Multicenter double-blind RCT (n = 4287)	Updated molnupiravir formulation was associated with 48% reduction in hospitalization or death compared to placebo, with non-inferior efficacy to nirmatrelvir/ritonavir and no observed mutagenic adverse events in 12-week follow-up	High
Lancet Public Health, 18 April 2026: *Performance of saliva-based rapid antigen tests for asymptomatic COVID-19 screening in community settings*	Prospective observational cohort (n = 12,456)	Saliva RATs had 89% sensitivity and 97% specificity for detection of asymptomatic Omicron XBB.3.1 infections when collected within 3 days of exposure	Moderate
ECDC Surveillance Report, 25 April 2026: *Effectiveness of voluntary vs mandatory masking policies in long-term care facilities during 2026 Q1 COVID-19 surge*	Retrospective ecological study (n = 2178 LTCFs across 12 EU countries)	Mandatory universal masking policies were associated with 32% lower COVID-19 outbreak rates and 41% lower COVID-19 related mortality rates compared to voluntary masking policies during periods of high community transmission	Low

2.3 Evidence Gaps and Future Research Needs

- Limited long-term (12+ month) safety data for next-generation bivalent COVID-19 vaccines in moderately to severely immunocompromised populations
- Lack of head-to-head RCT data comparing efficacy of available oral antivirals for newly emerging Omicron sublineages with significant spike protein mutations
- Insufficient empirical evidence on the causal impact of youth engagement in guideline development on adolescent/young adult guideline adherence and health outcomes
- No large-scale cost-effectiveness data for routine saliva RAT screening in workplace and educational settings during low community transmission periods

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Use nirmatrelvir/ritonavir as first-line oral antiviral for high-risk non-hospitalized patients; molnupiravir is only recommended as second-line for patients with contraindications to nirmatrelvir/ritonavir	New high-quality RCT demonstrates updated molnupiravir formulation has non-inferior efficacy to nirmatrelvir/ritonavir with no observed mutagenic risk	Use either nirmatrelvir/ritonavir or updated molnupiravir formulation as first-line oral antiviral options for high-risk non-hospitalized COVID-19 patients, with selection based on patient preference, drug interaction profile, and local supply availability

Rapid antigen tests are only recommended for diagnostic use in symptomatic individuals; PCR testing is required for asymptomatic screening	Moderate quality cohort data shows saliva RATs have high sensitivity and specificity for asymptomatic infection detection within 3 days of exposure	Saliva-based rapid antigen tests are recommended as an acceptable first-line option for asymptomatic COVID-19 screening following known exposure or during periods of high community transmission
Conditionally recommend voluntary masking in long-term care facilities during periods of elevated community transmission	Low quality population surveillance data shows mandatory masking is associated with significantly lower outbreak and mortality rates in LTCFs during surges	Strongly recommend mandatory universal masking for all staff, visitors, and residents in long-term care facilities during periods of high COVID-19 community transmission